MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER PRACTICAL EXAMINATION
(2024 ADMISION ONWARDS)

MG3DSCECT202 Internet of Things

Duration: 2 Hours **Maximum Marks: 35**

Model Question Paper

Instructions:

- **Demonstrate** the given task
- **Viva session** (Minimum 5 questions from the Practical Module)
- **Certified Lab Report**

Evaluation Criteria:

- **Viva**: 7 marks
- **Lab Report**: 8 marks
- **Demonstration**: 20 marks

Tasks(*Attempt any One* ***):***

1. With appropriate circuit diagram wire the system and execute the program to Read a button state with digital input using the ESP32/NodeMCU/ESP8266
2. With appropriate circuit diagram wire the system and execute the program to Dim an LED with PWM using the ESP32/NodeMCU/ESP8266
3. With appropriate circuit diagram wire the system and execute the program to Control a traffic light using ESP32/NodeMCU/ESP8266.
4. With appropriate circuit diagram wire the system and execute the program to set up a Basic Burglar alarm security system with the help of PIR sensor and buzzer using ESP32/NodeMCU/ESP8266.
5. With appropriate circuit diagram wire the system and execute the program for temperature sensor (DHT11) interfacing using ESP32/NodeMCU/ESP8266.
6. With appropriate circuit diagram wire the system and execute the program forto Bluetooth Interfacing using ESP32/NodeMCU/ESP8266.

7. With appropriate circuit diagram wire the system and Execute the program to Motor driver Interfacing using ESP32/NodeMCU/ESP8266.
8. Develop an appropriate interface and execute the program to create a simple web server in the ESP32.
9. With appropriate circuit diagram wire the system and execute the program to use ESP32 with ultrasonic sensor HC-SR04 to control servo motor.
10. With appropriate circuit diagram wire the system and Execute the program to Control LED matrix sign board via web interfacing using ESP32.